

Case Study: The High-Level Synthesis-Based Firmware Upgrade for the Digital Trigger Back-end System of the ATLAS Liquid Argon Calorimeter

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Abstract—The trigger and data acquisition systems of the LHC experiments rely on reconfigurable architectures based on FPGA (Field Programmable Gate Arrays) to accommodate design updates and increasing of the detectors complexity. However, this flexibility comes at the cost of longer development and validation cycles, particularly for large real-time systems operating under strict latency and throughput constraints. This work presents the HLS-based firmware redesign of the back-end system from the ATLAS Liquid Argon Calorimeter, where the introduction of a parallel switch matrix on REMAP block significantly simplifies the architecture and reduces the use of firmware resources. The proposed methodology allows systematic design exploration without modifying the source code, shortens verification time, and provides a more efficient workflow for creating and maintaining large-scale FPGA firmware in modern trigger systems.

Index Terms—HLS, Real-time Trigger Systems, FPGA.

I. INTRODUCTION

The ATLAS experiment at the Large Hadron Collider (LHC) processes collision data at 40 MHz, requiring real-time trigger decisions based on precise measurements from the Liquid Argon (LAR) calorimeter. Increasing luminosity and pile-up motivated the Phase-I upgrade, which introduced Super Cells (SCs) to improve granularity and data throughput, placing new demands on the trigger back-end firmware [1].

The LATOME board [2] implements this processing using FPGA-based firmware, but the legacy VHDL design relies on serialized data paths and multiple clock domain crossings, leading to high complexity and limited scalability. To address these limitations, this work proposes a redesign of the REMAP stage using High-Level Synthesis (HLS), enabling a simplified and more scalable architecture while maintaining strict latency and throughput constraints.

The main contributions of this work include the introduction of a parallel switch-matrix architecture for the REMAP stage, replacing serialized data processing; an HLS-based implementation that enables rapid design space exploration without modifying the functional description; and a quantitative evaluation demonstrating reductions in FPGA resource usage and verification time.

II. ARCHITECTURE REDESIGN

The legacy firmware processes data as serialized streams across multiple clock domains, requiring complex synchronization mechanisms and extensive control logic. In particular, the REMAP stage performs data reordering through sequential operations, introducing latency and limiting parallelism. Additionally, the USER processing stage requires 62 processing units, contributing significantly to FPGA resource usage [2].

The proposed redesign replaces this approach with a parallel architecture based on an Input Switch Matrix (ISM). The ISM is a configurable routing structure that maps input data streams directly to their required outputs in a single cycle. This eliminates time-multiplexed processing and reduces the need for intermediate buffering and synchronization.

The new architecture processes 384 input data words per cycle and produces 320 outputs, corresponding to the Super Cell organization required by the downstream processing. By operating entirely in parallel within a unified clock domain, the ISM simplifies the data path and prepares the system for further parallelization of subsequent processing stages.

III. HLS IMPLEMENTATION AND OPTIMIZATION

The ISM was implemented using HLS, enabling the description of routing logic at a higher level of abstraction. This approach allows rapid iteration and design exploration

through synthesis directives, without modifying the functional description. An initial implementation of the ISM used a loop-based multiplexer structure. Although functionally correct, it resulted in inefficient hardware synthesis due to undefined selection paths. An optimized version introduced explicit selection logic, ensuring deterministic mapping of all input cases and eliminating unnecessary inferred logic.

The impact of this optimization is summarized in Table I.

TABLE I
ISM OPTIMIZATION RESULTS

Version	Latency	ALMs	Fmax	Slack
HLS v0	1 cycle	26928	151 MHz	1.80 ns
HLS v1	1 cycle	21856	293 MHz	2.30 ns

Beyond functional optimization, HLS enabled systematic exploration of different architectural configurations, including sequential and combinational implementations. These configurations provide trade-offs between latency, resource usage, and timing performance.

TABLE II
ISM ARCHITECTURAL CONFIGURATIONS

Style	Latency	ALMs	Fmax
Sequential	2 cycles	21821	291 MHz
Combinational	1 cycle	21856	293 MHz

Loop unrolling was also used to control the degree of parallelism. A fully unrolled configuration instantiates one multiplexer per output stream, achieving single-cycle latency at the cost of higher resource utilization. Partial unrolling reduces resource usage but increases latency. Given the strict timing constraints of the trigger system [3], the combinational fully unrolled configuration was selected.

IV. INTERFACE AND INTEGRATION

Although the ISM operates on parallel data, the surrounding firmware modules remain serialized. To ensure compatibility, serializer-deserializer (SERDES) blocks were introduced at the REMAP interfaces. These blocks convert incoming serial streams into parallel data suitable for the ISM and serialize the outputs for downstream processing.

Clock domain crossing between the 320 MHz input domain and the 240 MHz processing domain is handled using a synchronization scheme based on a Start-of-Packet (SOP) signal. Dedicated logic ensures deterministic alignment of data during the transfer, avoiding metastability and preserving timing integrity.

V. VERIFICATION AND VALIDATION

The verification process employed multiple abstraction levels. Functional validation used an HLS C++ testbench for rapid iteration and early detection of design issues, followed by RTL verification via Cocotb to validate the full REMAP architecture, including SERDES and synchronization logic.

Hardware validation was then performed on a LATOME board under real operating conditions at CERN. The HLS testbench processed 10,000 events in 2.13 s versus 62.34 s for RTL, achieving a nearly $29\times$ speedup, which accelerates the development cycle and enables more extensive design space exploration.

VI. RESULTS AND DISCUSSION

The redesigned firmware achieves measurable improvements in both resource utilization and architectural simplicity. The number of processing units in the USER stage is reduced from 62 to 54, and overall FPGA logic usage decreases accordingly.

TABLE III
SYSTEM-LEVEL IMPROVEMENTS

Metric	Before	After	Change
Processing Units	62	54	-13%
ALMs	290k	270k	-7%
FPGA Utilization	69%	64%	-5 pp

Beyond quantitative improvements, the new architecture provides a cleaner and more modular design. The elimination of serialized processing and reduction of clock domain crossings simplify timing closure and improve maintainability. Furthermore, the use of HLS facilitates rapid adaptation to future requirements, enabling scalable evolution of the firmware.

VII. CONCLUSION

This case study demonstrates that High-Level Synthesis is an effective methodology for designing complex FPGA firmware in real-time systems. By introducing a parallel switch-matrix architecture in the REMAP stage, the LATOME firmware achieves reduced resource usage, improved timing performance, and significantly faster verification cycles.

The approach establishes a scalable framework for future upgrades, particularly the planned parallelization of additional processing stages. These results highlight the potential of HLS to address the growing complexity of trigger systems in high-energy physics while maintaining strict performance constraints.

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REFERENCES

- [1] Aleksa, M., Cleland, W., Enari, Y. "ATLAS Liquid Argon Calorimeter Phase-I Upgrade Technical Design Report." CERN Document Server, CERN-LHCC-2013-017, Geneva, 2013.
- [2] ATLAS Collaboration, "The Phase-I Trigger Readout Electronics Upgrade of the ATLAS Liquid Argon Calorimeters," JINST, P05003, 2022.
- [3] THE ATLAS HLT/DAQ/DCS GROUP. "ATLAS High-Level Trigger, Data Acquisition and Controls." ATLAS Technical Report, CERN, 2001.